Target conclusions about the one-loop correction Q_1 of the twisting supercharge, translated into the untwisted equivariant $\mathcal{N} = 4$ program

Step 2, directives R1–R4 (synthesis of the literature readers)

2026-10-28

1 Scope, labels, sources

Labels. [S] transcribed from the source at the cited equation (through the reader files below, which were checked against the arXiv PDFs); [D] derived in this file from quoted formulas (proof given or indicated); [I] identification or inference stated in no source. Nothing in this file is a computation of the program; every untwisted coefficient is a target to be computed.

Name clash (R3). Here Q_0, Q_1, Q_2 are the loop components of the twisted differential of the holomorphic-twist literature ([BGKWWY] (3.28), (D.13); [BK] (1.3), (1.6)). The manuscript's spinor parameters $Q_1 \in \Sigma$, $Q_2 \in \Sigma^*$ (Prop. prop:c7-susy) never appear: an element of Σ is written $\xi = (\xi_\alpha{}^i, \bar{\xi}_{\dot\alpha i})$ and an element of Σ^* is written $\eta = (\eta^\alpha{}_i, \bar{\eta}^{\dot\alpha i})$, as in `step2/notation/esw_to_weinberg_srednicki.tex` (Thm. thm:real). Spin-chain Q_k, S_k, H_k ([B03], [Z05]) are coefficients of g^k, $g^2 = g_{YM}^2 N/(8\pi^2)$, and are always written with the source key.

Inputs. `step2/literature/`: `sweep.json`, `gkw.tex/.json`, `bgkwwy.tex/.json`, `bgkwwy_checks_output.txt`, `n4_untwisted.tex/.json`; the dictionary `step2/notation/esw_to_weinberg_srednicki.tex`; the manuscript (ch02, ch06, ch07); DESIGN_STEP2.md, DEFINITIONS_v2.md; [ESW] read directly (local copy of arXiv:2002.10517, Defs. 2.16–2.18, Prop. 2.21, Rem. 1.29, §10.1.1, §10.3, §10.3.1, Thm. 10.9).

Two-component conventions (R1). Srednicki contractions $\chi\psi = \chi^\alpha\psi_\alpha$, $\bar\chi\bar\psi = \bar\chi_{\dot\alpha}\bar\psi^{\dot\alpha}$, $\varepsilon^{12} = +1$; Euclidean σ^m, $\bar\sigma^m$ of Def. def:wick of the dictionary; A_m, $A_{ij} = A_I\Sigma^I_{ij}$, $A^{ij} = \frac{1}{2}\varepsilon^{ijkl}A_{kl}$, $\lambda_\alpha{}^i$ $(S_+ \otimes W)$, $\bar\lambda_{\dot\alpha i}$ $(S_- \otimes W^*)$, antifields $A^*_m, A^*_{ij}, \lambda^{*\alpha}{}_i, \bar\lambda^{*\dot\alpha i}, c^*$; $(-,-)$ the coupling-carrying invariant pairing of $\mathfrak{g}$, contracting the $\mathfrak{g}$ factors in the order written. The unit is $\dfrac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}$, written out wherever it occurs.

Source keys.

[BK]	K. Budzik, J. Kulp, *Loop Corrected Supercharges from Holomorphic Anomalies*, arXiv:2512.07771v3 (JHEP 08 (2026) 089).
[BGKWWY]	K. Budzik, D. Gaiotto, J. Kulp, B. R. Williams, J. Wu, M. Yu, *Semi-Chiral Operators in 4d $\mathcal{N} = 1$ Gauge Theories*, arXiv:2306.01039v2 (JHEP 05 (2024) 245); PDF equation numbers.
[GKW]	D. Gaiotto, J. Kulp, J. Wu, *Higher Operations in Perturbation Theory*, arXiv:2403.13049v1.
[BGKWY22]	K. Budzik, D. Gaiotto, J. Kulp, J. Wu, M. Yu, *Feynman Diagrams in Four-Dimensional Holomorphic Theories and the Operatope*, arXiv:2207.14321v1.
[GWW]	Z. Gui, M. Wang, B. R. Williams, arXiv:2510.26608v2 (triangle weight only).

[W18] B. R. Williams, *Renormalization for holomorphic field theories*, arXiv:1809.02661v2.

[BGWW] P. Bomans, N. Garner, B. R. Williams, J. Wu, *Unravelling the Holomorphic Twist II*, arXiv:2509.16737v1.

[CLW] C.-M. Chang, Y.-H. Lin, J. Wu, *On $\frac{1}{8}$-BPS black holes and the chiral algebra of $\mathcal{N}=4$ SYM*, arXiv:2310.20086v2 (§3.1, (3.1), (3.17)–(3.19), footnote 5).

[CL25] J. Choi, E. Lee, *Konishi lifts a black hole*, arXiv:2511.09519v1.

[KvLM] T. Kibe, S. van Leuven, K. Mathieson, *Non-Coulomb Q-cohomology and quantum lifting*, arXiv:2610.22332v2.

[KKLP] S. Kim, S. Kim, S. Lee, J. Park, *Quantum black hole cohomologies*, arXiv:2606.27955v2.

[KLLS] S. Kim, J. Lee, S. Lee, J. Song, *Black hole cohomologies in $\mathcal{N}=4$ SYM*, arXiv:2610.20340v1.

[BMV] K. Budzik, H. Murali, P. Vieira, *Following Black Hole States*, arXiv:2306.04693v2 (abstract, main result, footnote 5).

[Z11] B. I. Zwiebel, *From Scattering Amplitudes to the Dilatation Generator in $\mathcal{N}=4$ SYM*, arXiv:1111.0083v2 (§2.3, (2.33)–(2.34)); with B. I. Zwiebel, *The psu(1,1|2) Spin Chain of $\mathcal{N}=4$ Supersymmetric Yang–Mills Theory*, Ph.D. thesis (its ref. [24]), not read.

[CY13] C.-M. Chang, X. Yin, *1/16 BPS States in $\mathcal{N}=4$ SYM*, arXiv:1305.6314v2.

[CL22] C.-M. Chang, Y.-H. Lin, *Words to describe a black hole*, arXiv:2209.06728v4.

[GGKM08] L. Grant, P. A. Grassi, S. Kim, S. Minwalla, arXiv:0803.4183v1.

[B03] N. Beisert, *The su(2|3) Dynamic Spin Chain*, hep-th/0310252v2.

[B03c] N. Beisert, *The Complete One-Loop Dilatation Operator of $\mathcal{N}=4$ SYM*, hep-th/0307015v5.

[B04] N. Beisert, dissertation, hep-th/0407277v4.

[Z05] B. I. Zwiebel, *$\mathcal{N}=4$ SYM to Two Loops: ... su(1,1|2) Sector*, hep-th/0511109v3.

[CDSW] F. Cachazo, M. R. Douglas, N. Seiberg, E. Witten, *Chiral Rings and Anomalies in Supersymmetric Gauge Theory*, hep-th/0211170v2.

[BKRS01] M. Bianchi, S. Kovacs, G. Rossi, Y. S. Stanev, hep-th/0104016v1.

[EJSS05] B. Eden, C. Jarczak, E. Sokatchev, Y. S. Stanev, hep-th/0501077v2 (regulator SSDR: statements only, never used as a regulator).

[ESW] C. Elliott, P. Safronov, B. R. Williams, arXiv:2002.10517 (project primary source).

2 The twisting supercharge as a generator of $\mathfrak{psl}(4|4)$ and the letter dictionary

2.1 The generator

[S] [ESW, Def. 2.16]: a square-zero supercharge is a nonzero $Q \in \Sigma$ with $\Gamma(Q,Q)=0$; it is holomorphic if the image of $\Gamma(Q,-)$ has dimension $n/2$. [ESW, §10.3, first bullet]: in 4d $\mathcal{N}=4$, "Rank $(1,0)$ and $(0,1)$ supercharges automatically square to zero. The corresponding twists are holomorphic." [ESW, §10.3.1]: the supercharge is $Q \in S_+ \otimes W$ of rank $(1,0)$ in the scalar summand spanning $\mathbb{C}w_1 \subset W$, with a homomorphism $\alpha : U(1) \to G_R$ giving Q weight 1. [BK] (before (3.51)): twisting supercharge $\mathcal{Q} := Q^4_-$; [CY13] (2.1): $Q = Q^4_-$, $S = S^-_4 = Q^\dagger$, $2\{Q, Q^\dagger\} = \Delta \equiv E - 2J - H_1 - H_2 - H_3$.

[D] In the project, the generator is

$$x = \xi = u \otimes w \in S_+ \otimes W, \quad u \in S_+,\ w \in W \text{ nonzero}; \qquad \xi_\alpha{}^i = u_\alpha w^i, \quad \bar\xi_{\dot\alpha i} = 0.$$

By (G1) of the dictionary $\Gamma(\xi,\xi)|_V = 0$ (it needs $\bar\xi$), and by (G2) $\Gamma_{kl}(\xi,\xi) = 2\varepsilon_{jikl} w^j w^i\, u_\alpha \varepsilon^{\alpha\beta} u_\beta = 0$: the

full ten-component $\Gamma(\xi,\xi)$ vanishes (dictionary, Thm. thm:real, last bullet; checked in `check_dictionary.py`, group "square-zero elements"). For $\bar\xi=0$ and a general ξ, $\Gamma_{kl}(\xi,\xi)=2\varepsilon_{jikl}\,\xi_\alpha{}^j\varepsilon^{\alpha\beta}\xi_\beta{}^i$ is ε_{jikl} contracted with the 2×2 minors of the 2×4 matrix $\xi_\alpha{}^i$; hence within $S_+\otimes W$ the ten-dimensional $\Gamma(\xi,\xi)$ vanishes exactly on the rank-one elements, whereas $[\xi,\xi]=\Gamma(\xi,\xi)|_V=0$ (Prop. prop:c2-special) for every $\xi\in S_+\otimes W$ (the rank-two ones being the topological twists of [ESW, §10.3, second bullet]). The rank-$(0,1)$ element $\bar\xi=\bar u\otimes n\in S_-\otimes W^*$ gives the conjugate twist and is treated identically with $S_\pm$, W, W^* exchanged.

[I] All rank-one ξ are related by $\mathrm{Spin}(4)\times SL(4)$ (orbit statement of [ESW, §2.5, §10.3]), so every target below is stated covariantly in (u,w); the matching of the label "Q^4_-" of [BK], [CY13] with a particular (u,w) in the basis of [SW, (17)] is a convention fixed in `step2/notation`. Index bookkeeping found here: in the project the gaugino and the supercharge parameter lie in the same Σ (the transformation is $-\Gamma(\xi,\lambda)$), whereas in [CY13], [GGKM08] the fermion $\Psi_{i\alpha}$ carries the index dual to that of Q^i_α; the project's upper W-index therefore corresponds to the lower $SU(4)$ index of the fermions and scalars of [CY13], [GGKM08] (e.g. Φ^{4I} of [BK] (3.51) $\leftrightarrow A_{ij}w^j$ below).

Throughout, fix $u'\in S_+$ with $u'^\alpha u_\alpha=1$ and $n\in W^*$ with $n_iw^i=1$; then $\lambda_\alpha{}^i=(u'\lambda^i)\,u_\alpha-(u\lambda^i)\,u'_\alpha$ with $(u'\lambda^i)=u'^\alpha\lambda_\alpha{}^i$, $(u\lambda^i)=u^\alpha\lambda_\alpha{}^i$ [D].

2.2 The equivariant vertices along ξ

[D] Substituting $\xi_\alpha{}^i=u_\alpha w^i$, $\bar\xi=0$ into (F1), (F3) of the dictionary (= ch02 eq:c2-S1Q, eq:c2-S2), using $u_\alpha\varepsilon^{\alpha\beta}=-u^\beta$ and $\varepsilon^{\alpha\beta}u_\beta=u^\alpha$:

$$S^{(1)}(\xi)=\int\Big\{-i\big(\bar\lambda_{\dot\alpha i}w^i\,\bar\sigma^{m\dot\alpha\alpha}u_\alpha,\,A^*_m\big)+\tfrac12\varepsilon_{jikl}w^j\big((u\lambda^i),A^{*kl}\big)-\tfrac14\big(F_{mn}(\sigma^m\bar\sigma^n u)_\alpha,\,\lambda^{*\alpha}{}_iw^i\big)$$
$$-\tfrac14\big([A^{ik},A_{kj}w^j]u_\alpha,\,\lambda^{*\alpha}{}_i\big)-\tfrac{i}{2}\big(u^\alpha\sigma^m_{\alpha\dot\alpha}D_m(A_{ij}w^j),\,\bar\lambda^{*\dot\alpha i}\big)\Big\},$$

$$S^{(2)}(\xi,\xi)=-\tfrac12\int\big(u_\alpha w^i\lambda^{*\alpha}{}_i,\;u_\beta w^j\lambda^{*\beta}{}_j\big),$$

the two other terms of (F3) vanishing because $\Gamma(\xi,\xi)=0$ (this confirms inference I2 of `gkw.tex`). With the sign pattern $\xi.A=-\Gamma(\xi,\lambda)$, $\xi.\lambda=-\frac12\rho(F_A)\xi$ (dictionary, Cor. cor:closure):

$$\xi.A_m=-i\,(\bar\lambda_{\dot\alpha i}w^i)\,\bar\sigma^{m\dot\alpha\alpha}u_\alpha,\qquad \xi.A_{kl}=2\varepsilon_{jikl}w^j(u\lambda^i),$$
$$\xi.\lambda_\alpha{}^i=\tfrac14F_{mn}(\sigma^m\bar\sigma^nu)_\alpha w^i+\tfrac14[A^{ik},A_{kj}w^j]\,u_\alpha,\qquad \xi.\bar\lambda_{\dot\alpha i}=\tfrac{i}{2}\,u^\alpha\sigma^m_{\alpha\dot\alpha}D_m(A_{ij}w^j),\qquad \xi.c=0.$$

2.3 Letters

[D] Consequences of the last display: $\xi.(A_{kl}w^l)=0$ ($\varepsilon_{jikl}w^jw^l=0$); $\xi.(\bar\lambda_{\dot\alpha i}w^i)=0$ ($A_{ij}w^iw^j=0$); the scalars A_{kl} transverse to w map onto $(u\lambda^i)$ modulo w; modulo w^i, $\xi.(u'\lambda^i)=\frac14[A^{ik},A_{kj}w^j]$ and $\xi.(u\lambda^i)=0$; with $w=e_4$, $A^{ik}=\varepsilon^{ikl}A_{l4}$ for $i,k\le3$, so $\xi.(u'\lambda^i)\propto\varepsilon^{ikl}[A_{l4},A_{k4}]$, the structure of [CY13] (2.5) $\{Q,\psi_n\}=-i\epsilon_{nmp}[\phi^m,\phi^p]$ and of [BK] (3.50) $Q_0\beta_I\ni\frac12\varepsilon_{IJK}[\gamma^J,\gamma^K]$. On the w-component of λ, $u^\alpha\,\xi.\lambda_\alpha=\frac14u^\alpha F_{mn}(\sigma^m\bar\sigma^n)_\alpha{}^\beta u_\beta$ and $u'^\alpha\,\xi.\lambda_\alpha$ contains $\frac14u'^\alpha F_{mn}(\sigma^m\bar\sigma^n)_\alpha{}^\beta u_\beta$; the curvature component not reached is $u'^\alpha F_{mn}(\sigma^m\bar\sigma^n)_\alpha{}^\beta u'_\beta$. The derivatives $u^\alpha\sigma^m_{\alpha\dot\alpha}D_m$ of $A_{ij}w^j$ are reached $(\xi.\bar\lambda)$, in agreement with [BK] (2.5) ($\{\mathcal{Q},\bar Q^1_{\dot\alpha}\}=P_{-\dot\alpha}$, antiholomorphic translations exact).

twisted letter	[BK] (3.51) [S]	project, two-component [I]	project, index-free [I]	(weight, ghost)
γ	$\gamma^I\sim\Phi^{4I}$	$A_{ij}w^j$ (3 components, $(A_{ij}w^j)w^i=0$)	contraction with w of $A|_{W^\perp_{\mathbb{R}}}$ read in $\wedge^2W^*$	$(1,0)$
β	$\beta_I\sim\Psi_{I+}$	$(u'\lambda^i)=u'^\alpha\lambda_\alpha{}^i$ modulo w^i	coefficient of u in $\lambda|_{S_+\otimes W}$, read in $W/\mathbb{C}w$	$(\frac32,0)$
$\partial_{\dot\alpha}c$	$\partial_{\dot\alpha}c\sim\bar\Psi^4_{\dot\alpha}$	$\bar\lambda_{\dot\alpha i}w^i$	$\lambda|_{S_-\otimes W^*}$ paired with w	$(\frac32,0)$

c	ghost	c	c	$(0,1)$
b	$b \sim F_{++}$	$u'^{\alpha} F_{mn} (\sigma^m \bar{\sigma}^n)_{\alpha}{}^{\beta} u'_{\beta}$ (normalisation open); alternative representative: a component of λ^*	F_A self-dual part evaluated on $u' \otimes u'$	$(2,0)$; $(\frac{5}{2}, -1)$
$\partial_{\dot\alpha}$	$\partial_{\dot\alpha} \sim D_{+\dot\alpha}$	$u'^{\alpha} \sigma^m_{\alpha\dot\alpha} D_m$ (exact directions: $u^{\alpha} \sigma^m_{\alpha\dot\alpha} D_m$)	covariant derivative along $u' \otimes S_- \subset V_{\mathbb{C}}$	$(1,0)$

Remarks. (i) [BK] (3.51) and [BGKWWY] §3 (bullets after (3.3)) state the dictionary with "$\sim$", up to field redefinitions at the level of free $\mathcal{Q}$-cohomology and without normalisation factors [S]. (ii) [ESW, Thm. 10.9 and the collections after its proof] [S]: the twisted fields are $c + A_{0,1} + B$, $\varphi + \psi + \varsigma^*$, $\xi_{\mathrm{ESW}} + b^* + a^*$, $a + b + \xi^*_{\mathrm{ESW}}$, $\varsigma + \psi^* + \varphi^*$, $B^* + A^*_{0,1} + c^*$ (ESW's own symbols; ξ_{ESW} is ESW's $\Omega^{2,0}$ gauge fermion); untwisted antifields occur in the twisted fields, and the $(2,0)$-form part of the twisted b-field is the antifield B^*. Hence a twisted letter may be represented by an untwisted field or by an antifield; the two representatives of b in the table differ by one unit of ξ (Lemma 1). (iii) The choice of u' is immaterial modulo exact terms: $u' \to u' + tu$ adds $t(u\lambda^i)$, which is ξ.-exact modulo w [D].

2.4 From the equivariant cochain to the twisted differential

[S] [ESW, Rem. 1.29]: an action of $\mathfrak{g}$ is $I_{\mathfrak{g}} = I + \sum_{k\geq 1} S^{(k)}_{\mathfrak{g}}$ with $(d_{CE} + Q_{BV}) I_{\mathfrak{g}} + \frac{1}{2}\{I_{\mathfrak{g}}, I_{\mathfrak{g}}\} = 0$. [ESW, Def. 2.17]: the Q-twisted theory has the same fields and pairing and action $S^Q = S + \sum_{k\geq 1} S^{(k)}_{\mathcal{A}}(Q,\dots,Q)$. [ESW, Def. 2.18]: for α compatible with Q, $S^{(k)}_{\mathcal{A}}(Q,\dots,Q)$ has α-weight k and cohomological degree $-k$, so S^Q has total degree 0. [ESW, Prop. 2.21]: twisting commutes with dimensional reduction. Manuscript Def. def:c7-component: $O[\infty] = \lim_{\varepsilon\to 0} \partial_\delta W(P(\varepsilon,\infty), I_{\mathfrak{psl}(4|4)} - I^{CT}_{\mathfrak{psl}(4|4)}[\varepsilon] + \delta O(\varepsilon))$, components $[(d_{CE}+Q)O[\infty]](x_1)$ with local representatives $\int \omega\, f_{x_1}$; Lemma lem:c7-image: the arity-one component is Q of the arity-one part of $O[\infty]$.

Definition 1 (evaluation along ξ, [D])**.** Since $[\xi,\xi] = 0$, $\mathbb{C}\xi$ is an abelian sub-superalgebra; restriction of cochains to it is a chain map $C^\bullet(\mathfrak{psl}(4|4)) \to C^\bullet(\mathbb{C}\xi)$, and $C^\bullet(\mathbb{C}\xi)$ has zero differential. Write $\mathrm{ev}_\xi F = \sum_k c_k F^{(k)}(\xi,\dots,\xi)$ for the resulting evaluation, the constants c_k fixed by the normalisation of cochain values in which [ESW, Def. 2.17] carries coefficient 1.

Then, whenever the one-loop equivariant quantum master equation holds along ξ and (ξ,ξ) after finite counterterms (Step 2), the twisted quantum differential on observables is $\mathrm{ev}_\xi \circ (d_{CE} + Q)$ [I], with order-$\hbar$ part

$$\underbrace{Q\, O[\infty]\big|_{\hbar}}_{\text{arity 0}} + c_1 \underbrace{[(d_{CE}+Q)O[\infty]](\xi)\big|_{\hbar}}_{\text{arity one}} + c_2 \underbrace{[(d_{CE}+Q)O[\infty]](\xi,\xi)\big|_{\hbar}}_{\text{arity two}} + \cdots,$$

and the arity-two component on (ξ,ξ) equals Q of the arity-two part of $O[\infty]$, because d_{CE} of the arity-one part evaluated on (ξ,ξ) is the arity-one part on $[\xi,\xi] = 0$ [D]. The twisted Q_1 on twisted letters is the order-$\hbar$ part of this differential transported to the twisted letters by the homotopy equivalence of [ESW, Thm. 10.9] (elimination of trivial pairs, [ESW, proof of Thm. 10.1, Props. 1.21, 1.23]); in [BGKWWY] (D.13)–(D.15) form, $\tilde{Q} = \pi_0(\hbar Q_1 + \cdots)\Delta^{-1}\iota_0$ [I].

Lemma 1 (arity count, [D])**.** Let $O = \int \omega\, f$ have Weyl weight w_O and ghost degree $|O|$. The arity-k component on $(\xi,\dots,\xi)$ has local representatives of Weyl weight $w_O + \frac{k}{2}$ and ghost degree $|O| + 1 - k$. Hence a twisted pair entry with untwisted representatives of weights and ghost degrees $(w_{\mathrm{in}}, g_{\mathrm{in}}) \mapsto (w_{\mathrm{out}}, g_{\mathrm{out}})$ lies in arity

$$k = 2(w_{\mathrm{out}} - w_{\mathrm{in}}) = 1 + g_{\mathrm{in}} - g_{\mathrm{out}},$$

and the two expressions must agree (a check of the dictionary). For outputs without antifields ($g_{\mathrm{out}} \geq 0$), $k \leq |O| + 1$.

Proof. Prop. prop:c7-select: each value of the conformal Killing spinor carries weight $-\frac{1}{2}$, and each evaluation on a generator (ghost degree 1) lowers the ghost degree by one; $(d_{CE}+Q)O[\infty]$ has degree $|O|+1$. Apply this k times. □

Lemma 2 (where $S^{(2)}$ enters, [D])**.** $S^{(2)}(\xi,\xi)$ has only λ^*-legs. $P(\varepsilon,\infty)$ has components in the slots $A\otimes A$, $c\otimes A^*$, $A^*\otimes c$, $\lambda\otimes\lambda$ only (dictionary, Prop. prop:prop), so a λ^*-leg is never on a $P(\varepsilon,\infty)$ edge. In the expansion of $(d_{CE}+Q)O[\infty]$ by the UV-boundary identity (ch07 eq:c7-special, eq:c7-defect) every graph has at most one field–antifield pairing (a $K_\varepsilon - K_0$ edge, a K_ε self-contraction, or the classical bracket $\{I_{\mathfrak{psl}(4|4)}, O\}$), the other edges being $P(\varepsilon,\infty)$. A connected graph containing the vertex $S^{(2)}(\xi,\xi)$ therefore joins it through that single pairing, and its second λ^*-leg is external; for the same reason no one-loop counterterm along (ξ,ξ) comes from a graph with an $S^{(2)}$ vertex. Consequently $S^{(2)}(\xi,\xi)$ contributes at one loop only to outputs containing λ^*; field-only outputs of the (ξ,ξ) component come from graphs with two $S^{(1)}(\xi)$ vertices. $S^{(2)}$ enters the field-only data through the (ξ,ξ) component of eq:c2-mc (lem:c2-components: $\{S, S^{(2)}(\xi,\xi)\} + (-1)^{|\xi|}\{S^{(1)}(\xi), S^{(1)}(\xi)\} = 0$) and through twisted letters represented by antifields (Remark (ii) above).

2.5 Loop order and coupling

[S] [BK] (1.3), (2.22): $\hbar$ counts loops, the twisted actions (3.1), (3.53) carry no coupling. [KKLP] (2.13): $\mathcal{Q} = \mathcal{Q}_{\text{free}} + g_{YM}\mathcal{Q}_0 + g_{YM}^2\mathcal{Q}_1 + O(g_{YM}^3)$, and $\mathcal{Q}_0 + g_{YM}^2\mathcal{Q}_1$ after $\Phi \to g_{YM}^{-1}\Phi$ (2.2); [KLLS] footnote 2 the same. Project: the coupling sits in $(-,-)$, $(X,Y) = -\frac{4\hbar}{g^2}\operatorname{tr}(XY)$ for $\mathfrak{su}(N)$ (dictionary, Cor. cor:wick), $\dfrac{\hbar\,\operatorname{tr}_{\mathfrak{g}}(\operatorname{ad}_X\operatorname{ad}_Y)}{16\pi^2(X,Y)} = -\frac{g^2N}{32\pi^2}$ (ch06 rem:c6-konishi). [D] [`n4_untwisted` D1]: an $n \to m$ letter term at order g^k (canonical normalisation) has $L = (k-m+n)/2$ loops. The twisted Q_1 (two letters to two letters, order g_{YM}^2) is one-loop, i.e. order $\hbar$ in the project; the twisted Q_0 (one letter to two, order g_{YM} canonically) is classical. In spin-chain language the twisted Q_1 has the leg count of [B03]'s order-g^2 Q_2 (two sites to two), not of [B03]'s order-g Q_1 (`sweep.json`, leg-counting dictionary, [I]).

3 Target conclusions

Each target lists (a) source statement, (b) conventions, (c) untwisted translation, (d) scheme status. "Pair value" means the representative of Q_1 on a pair of letters; "class" means the induced map on $H(Q_0)$.

Target 1 (structure of the loop expansion)**.** (a) [S] $Q_n\mathcal{O} = \frac{1}{(n+1)!}\{\mathcal{I},\dots,\mathcal{I},\mathcal{O}\}$ ([BGKWWY] (3.33); [BK] (1.6)); $\mathcal{Q}\mathcal{O} = \{I,\mathcal{O}\}_0 + \frac{\hbar}{2!}\{I,I,\mathcal{O}\}_0 + \cdots$ ([BK] (2.22)–(2.24)); $\sum_{n_1+n_2=n} Q_{n_1}Q_{n_2} = 0$, $\{Q_0,Q_1\} = 0$, $Q_1^2 + \{Q_0,Q_2\} = 0$ ([BK] (2.25), (2.26), (1.12); [BGKWWY] (3.41)–(3.42)); $Q_1^2 \neq 0$, hence $Q_2 \neq 0$ ([BGKWWY] §3.2.3; (C.8)–(C.9) for $SU(2)$). [GKW] (2.27): $QO = \{O\} + \{I,O\} + \frac{1}{2}\{I,I,O\} + \cdots$; (3.10) is the global constraint $n|\Gamma_0| = (n-1)|\Gamma_1| + n + 1$ on n-Laman graphs. [BK] after (2.43): "at one-loop there is only the triangle diagram". [KvLM] (5.1)–(5.2): $Q_0^2 = 0$, $\{Q_0,Q_1\} = 0$, $Q_1^2 = -\{Q_0,Q_2\}$, so $[Q_1]$ is nilpotent on $H(Q_0)$.

(b) Twisted, $\hbar$ loop counting, regulator: Schwinger time $t \geq \epsilon$ on every propagator, finite without counterterms ([BK] (2.42), footnote 22; [GKW] App. B; [W18] Thm. 3.4).

(c) [I] $Q_0 \leftrightarrow$ the order-$\hbar^0$ part of $\mathrm{ev}_\xi\circ(d_{CE}+Q)$: $Q + \{I + S^{(1)}(\xi) + c_2 S^{(2)}(\xi,\xi), -\}$ on classical observables. $Q_1 \leftrightarrow$ the order-$\hbar$ part: components along ξ (arity one) and (ξ,ξ) (arity two). $\{Q_0,Q_1\} = 0 \leftrightarrow$ the order-$\hbar$ part of the (ξ) and (ξ,ξ) components of $(d_{CE}+Q)^2O[\infty] = 0$. $Q_1^2 = -\{Q_0,Q_2\}$ is an order-$\hbar^2$ statement (Step 5); its one-loop content is $[Q_1]^2 = 0$ on $H(Q_0)$. The untwisted one-loop graphs (at most one field–antifield pairing, the other edges $P(\varepsilon,\infty)$) are not only triangles: bubbles and self-energy insertions on one leg of O also occur; they must reduce to the terms of Section 4.

(d) Scheme independent: $\{Q_0,Q_1\} = 0$; $[Q_1]^2 = 0$ on $H(Q_0)$. Scheme dependent: "triangle only", and the letter count $n+1 \to 2$ of [BGKWWY] (3.44)–(3.45).

Target 2 (action on pairs; single letters)**.** (a) [S] [BK] (3.7)–(3.8): $Q_1(f_1\cdots f_n)=\sum_{i<j}(-1)^{\epsilon_{ij}}Q_1(f_if_j)\,f_1\cdots\hat{f}_i\cdots\hat{f}$ $\epsilon_{ij}=|f_i|\sum_{k<i}|f_k|+|f_j|\sum_{k<j,k\neq i}|f_k|$. [BGKWWY] (C.1)–(C.2): $Q_1\mathcal{O}=\frac{1}{2}\{\mathcal{I},\mathcal{I},\mathcal{O}\}$ vanishes on single fields, acts on pairs; (3.43): $Q_nV(N_b,N_c,J_1,J_2)=0$ if $N_b+N_c<n+1$.

(b) [BK] (3.9)–(3.10):

$$\mathcal{D}^{\triangle}_{w,z}(f,g)=\sum_{m,n}\frac{1}{(n+m+2)!}e^{z\cdot\partial}((w-z)\cdot\partial)^n\partial_{\dot\alpha}f\;e^{w\cdot\partial}((z-w)\cdot\partial)^m\partial^{\dot\alpha}g\Big|_{y=0},\qquad \mathcal{D}^{\triangle}_{0,0}(f,g)=\tfrac{1}{2}\partial_{\dot\alpha}f\partial^{\dot\alpha}g.$$

(c) [I] Pair rule $\leftrightarrow$ one-loop graphs whose loop meets exactly two legs of O. Single letters: Step-1 result $O(\varepsilon)=O$ for every letter (DEFINITIONS_v2); target: the order-$\hbar$ component along ξ on $\int\omega\,(X,A_{ij}w^j)$, $\int\omega\,(X,u'\lambda^i)$, $\int\omega\,(X,\bar\lambda_{\dot\alpha i}w^i)$, $\int\omega\,(X,c)$ and on the F-letter is of the form of Section 4 (multiples of the classical component and $[Q_0,\varphi_1]$).

(d) Scheme dependent [D]: a finite coefficient y in front of $\hbar\,s\big(2\int(A,A^*)-4\int(\lambda^*,\lambda)-\int(c^*,c)\big)$ (the direction of the logarithmic counterterm, thm:c4-log) is a rescaling N_y of the letters by their field type and shifts $Q_1\mapsto Q_1+[Q_0,N_y]$; on β_I, with $Q_0\beta_I=[c,\beta_I]+\frac{1}{2}\varepsilon_{IJK}[\gamma^J,\gamma^K]$ ([BK] (3.50)), this is the one-letter-to-two-letter term $\pm\big(y_c[c,\beta_I]+(2y_\gamma-y_\beta)\frac{1}{2}\varepsilon_{IJK}[\gamma^J,\gamma^K]\big)$ (y_c, y_γ, y_β the rescaling exponents of c, γ, β), nonzero for generic y. The vanishing on single letters and the pairwise form hold in the counterterm-free twisted scheme ([BK] footnote 22, [W18] Thm. 3.4) and are not scheme-independent statements.

Target 3 (vanishing on the c,γ letters; chiral primaries)**.** (a) [S] [BK] (3.11): $Q_1(cc)=Q_1(\gamma\gamma)=Q_1(\gamma c)=Q_1(\beta c)=0$. [BK] §3.4.1, (3.77): $Q_1(\gamma^I\gamma^J)=0$, hence Q_1 vanishes on the representatives $\mathrm{Tr}\,\gamma^{(I_1}\gamma^{I_2}\cdots\gamma^{I_n)}$ at any N, and their $\mathfrak{psl}(3|3)$ descendants are Q_1-closed (§3.4.1, using footnote 24); the single-trace$\to$single-trace part does not lift the infinite-N single-trace cohomology; the trace-splitting/joining part is left open.

(b) Propagators pair $c\leftrightarrow b$, $\gamma\leftrightarrow\beta$ ([BK] (2.31)–(2.33), (3.1)); vertices $\mathrm{Tr}\,b[c,c]$, $\beta[c,\gamma]$, $W[\gamma]=\mathrm{Tr}\,\gamma^1[\gamma^2,\gamma^3]$ ([BK] (3.48)).

(c) [D] Wick count: in the triangle each field of the pair is contracted into one of the two vertices and the third edge joins the two vertices; for a pair drawn from $\{\partial^k c,\partial^k\gamma\}$ the remaining legs of the two vertices contain no b and no β, so no third edge exists. Hence Q_1 vanishes on every polynomial in the derivatives of c and γ, in particular on the letters $\{\gamma,\partial_{\dot\alpha}c\}$ that [`n4_untwisted` D8] identifies with the content of the su(2|3) sector [I].

[D] Untwisted, $n=2$: for $w_1,w_2\in W^\perp_{\mathbb{R}}\otimes\mathbb{C}$ with $w_{a\,ij}w^j=0$ ($a=1,2$; the three-dimensional plane of w), $\iota_{w_a}A=\frac{1}{4}w_a^{ij}A_{ij}$ is a combination of the components $A_{ij}w^j$ and $(w_1,w_2)=0$; so $\mathrm{Tr}\,\gamma^{(I}\gamma^{J)}\leftrightarrow O=\int\omega\,(\iota_{w_1}A,\iota_{w_2}A)$ in the traceless part (Step-1 dilatation coefficient 0). By Cor. cor:c7-Q, for $(w_1,w_2)=0$ the component along ξ is a multiple, with an (ξ,w_1,w_2)-independent coefficient, of

$$\int\omega\Big[(\iota_{w_1}A,(\rho(w_2)\xi,\lambda))+(\iota_{w_2}A,(\rho(w_1)\xi,\lambda))\Big],$$

and $(\rho(w_a)\xi,\lambda)=\iota_{w_a}\Gamma(\xi,\lambda)=-\frac{1}{2}w_a^{kl}\varepsilon_{jikl}w^j(u\lambda^i)=0$ for w_a in the plane of w. Hence the component along ξ of $\int\omega\,(\iota_{w_1}A,\iota_{w_2}A)$, w_1,w_2 in the plane of w, vanishes at every order at which the hypotheses of Prop. prop:c7-select hold: the untwisted reproduction of [BK] (3.77) for $n=2$, exactly and not only in cohomology. Two-component form: $O=\int\omega\,(A_{ij}w^jm_1^i,\,A_{kl}w^lm_2^k)$, $m_1,m_2\in W$ [D]. For $n\geq3$ ($\int\omega$ symmetrised tr of products of $A_{ij}w^j$) the corresponding statement is a Step-4 target.

(d) Pair values: scheme dependent in general; for the $n=2$ chiral primary the exact vanishing is a consequence of the selection rule and holds in every equivariant scheme. Class level ($[Q_1]$ annihilates $\mathrm{Tr}\,\gamma^{(I_1}\cdots\gamma^{I_n)}$ and descendants): scheme independent.

Target 4 (the bc entry)**.** (a) [S] [BK] (3.64): $Q_1(b^Ac^B)=\frac{1}{2}f_{ACD}f_{BCE}\,\partial_{\dot\alpha}c^D\partial^{\dot\alpha}c^E$; at nonzero shift $Q_1(b^A(z)c^B(w))=f_{ACD}f_{BCE}\mathcal{D}^{\triangle}_{w,z}(c^D,c^E)$ ((3.13), (3.21)); with derivatives (B.5): $Q_1(b^A\partial_1^m\partial_2^nc^B)=f_{ACD}f_{BCE}\frac{1}{m+n+2}\sum_{k,l}\frac{1}{k+l+1}\binom{m}{k}\binom{n}{l}\,\partial_1^k\partial_2^l\partial_{\dot\alpha}c^D\,\partial_1^{m-k}\partial_2^{n-l}\partial^{\dot\alpha}c^E$; footnote 23 thanks J. Park for pointing out a missing factor $\frac{1}{2}$ in the formulas for $z=w=0$ in Sections 3.2–3.4. [BGKWWY] (C.4) ($SU(2)$):

$Q_1(b_A c^B) = -(\partial_\alpha c^A \partial^\alpha c^B - \delta^B_A \partial_\alpha c_D \partial^\alpha c^D)$, equal to $f_{ACE} f_{BDE} \partial_\alpha c^C \partial^\alpha c^D$ (`bgkwwy_checks` K1/K2). [D] At face value with $f = \varepsilon$, [BGKWWY] (C.4) = 2×[BK] (3.35)/(3.64) and [BGKWWY] (C.3) = 2×[BK] (3.36), with the same index structure; [I] this factor 2 is the factor $\frac{1}{2}$ of [BK] footnote 23.

(b) $[t_A, t_B] = f_{AB}{}^C t_C$, indices moved with δ^{AB}, Killing normalisation ([BK] (3.12)); $X \cdot Y = \partial_{\dot\alpha} X \partial^{\dot\alpha} Y$; [BK] (3.1) has $+\frac{1}{2}[c,c]$, [BGKWWY] (3.2) has $-\frac{1}{2}[c,c]$; [BGKWWY] App. C has internal factor-2 and sign discrepancies ((C.5) vs (C.6)–(C.7); `bgkwwy_checks` CK03, CK04).

(c) [D] Colour: $\frac{1}{2} f_{ACD} f_{BCE} X^A Y^B Z^D Z'^E = \frac{1}{2}([X,Z],[Y,Z'])_\delta$ with $(-,-)_\delta$ the δ^{AB} pairing, $\mathrm{tr}(\mathrm{ad}_Z \mathrm{ad}_{Z'}) = -f_{ACD} f_{ACE} Z^D Z'^E$. Untwisted operator (open colour $X, Y \in \mathfrak{g}$): $O = \int \omega \left(X, u'^\alpha F_{mn} (\sigma^m \bar\sigma^n)_\alpha{}^\beta u'_\beta\right)(Y, c)$, $w_O = 2$, $|O| = 1$. Output (field-only): $\int \omega \left([X, \bar\lambda_{\dot\alpha i} w^i], [Y, \bar\lambda^{\dot\alpha}{}_j w^j]\right)$, weight 3, ghost 0. Lemma 1: $k = 2(3-2) = 2 = 1 + 1 - 0$: the (ξ,ξ) component at order $\hbar$; by Lemma 2 its field-only part comes from graphs with two $S^{(1)}(\xi)$ vertices, one O vertex and one K_ε edge. With b represented by its antifield component ([ESW] Thm. 10.9 collections), the same entry is an arity-one component ($k = 2(3-\frac{5}{2}) = 1 = 1 + 0 - 0$) [D].

(d) Pair value scheme dependent (Section 4). Class level: see Target 9.

Target 5 (the $\beta\gamma$ entry: Konishi)**.** (a) [S] [BK] (3.66): $Q_1((\beta_I)^A(\gamma^J)^B) = \frac{1}{2}\delta^J_I f_{ACD} f_{BCE}\, \partial_{\dot\alpha} c^D \partial^{\dot\alpha} c^E$; general representation (3.16): $Q_1((\beta_I)_i(z)(\gamma^J)^j(w)) = -\delta^J_I (T_B T_A)^j{}_i \mathcal{D}^\triangle_{w,z}(c^A, c^B)$. [CL25] (2.5)–(2.6): $Q_K = \bar A_{ij,kl} \frac{g^2_{YM}}{32\pi^2} (\partial_\psi)_{ij} (\partial_{\bar\Phi})_{kl}$ (SO(N); lower indices, $A_{ij,kl}$ defined in (2.6)), the correction "inferred" from the generalised Konishi anomaly, with "other quantum corrections to the supercharge" possible for more complicated composite operators (after (2.6), §4); unnumbered example after (2.6): $Q \,\mathrm{Tr}\, \bar\Phi \psi_+ = g_{YM} \,\mathrm{Tr}\, \bar\Phi \frac{\partial \bar W}{\partial \bar\Phi} + g^2_{YM} \frac{N-2}{16\pi^2} \,\mathrm{Tr}\, \bar\lambda_{\dot\alpha} \bar\lambda^{\dot\alpha}$; from the generalised Konishi anomaly (2.1)–(2.2), $\bar D^2 \,\mathrm{Tr}\, \bar\Phi e^{[V,\cdot]} \Phi|_{\text{anomaly}} = g^2_{YM} \frac{C(\mathrm{adj})}{16\pi^2} \,\mathrm{Tr}\, W_\alpha W^\alpha$. [KvLM] (5.5): $Q_1\big((\psi_m)^a(\phi^n)^b\big) = -\kappa_Q\, \delta^n_m\, f^a{}_{cd} f^{bc}{}_e (\lambda_{\dot\alpha})^d (\lambda^{\dot\alpha})^e$, with κ_Q "the non-zero normalization … whose precise value will not be needed" (after (5.5)), a fixed nonzero constant left uncomputed; (5.21): $[Q_1][F_4] = -12\kappa_Q[\mathcal{S}^4]$. [KKLP] (2.7) ($\frac{1}{4}\kappa^2$, $\mathrm{tr}(T_A T_B) = \kappa \delta_{AB}$, (2.4)); [KLLS] (2.4) ($\kappa_1^2/16$). Untwisted: [CDSW] (3.3): $\bar D^2 J = \frac{1}{32\pi^2} \,\mathrm{tr}_{\mathrm{adj}}(\mathrm{ad} W_\alpha)(\mathrm{ad} W^\alpha)$; (3.4), (3.14). [BKRS01] (7): $\frac{1}{4}\bar D^2 : \mathrm{tr}(e^{-2gV} \Phi^\dagger_I e^{2gV} \Phi^I) := \mathrm{tr}(\Phi^I \partial \mathcal{W}/\partial \Phi^I) + \frac{6g^2N}{32\pi^2} \,\mathrm{tr}(W^\alpha W_\alpha)$; (9): $(\mathcal{K}_{\mathbf{10}^*})_{AB} = 3\sqrt{2} g\, t^{ijk}_{AB} \,\mathrm{tr}(\varphi_i \varphi_j \varphi_k) + \frac{6g^2N}{32\pi^2} \,\mathrm{tr}(\bar\psi_{\dot\alpha A} \bar\psi^{\dot\alpha}_B)$. [EJSS05] (1): $\bar D^2 K = B + \frac{g^2}{32\pi^2} F$; (8) one loop: $\bar D^2 K = \frac{g}{6} \epsilon_{IJK} \,\mathrm{Tr}(\Phi^I [\Phi^J, \Phi^K]) + \frac{g^2 N}{32\pi^2} \,\mathrm{Tr}(W^\alpha W_\alpha)$; (12): $Z_F = \frac{g^2 N}{32\pi^2}$.

(b) [CL25]: $\tau = \frac{\theta}{2\pi} + \frac{4\pi i}{g^2_{YM}}$, $\mathrm{Tr}\, T^A_R T^B_R = C(R)\delta^{AB}$; [EJSS05]: $\mathrm{Tr}(T^a T^b) = \delta^{ab}$ (after (109)); (109): $S_{\mathcal{N}=4} = \int d^4x d^2\theta d^2\bar\theta \,\mathrm{Tr}\, e^{gV} \bar\Phi_I e^{-gV} \Phi^I + \frac{1}{4} \int d^4x_L d^2\theta \,\mathrm{Tr}\, W^\alpha W_\alpha + \frac{g}{3!} \int d^4x_L d^2\theta\, \epsilon_{IJK} \,\mathrm{Tr}\, \Phi^I [\Phi^J, \Phi^K] + \text{c.c.}$; propagators (110), (112); (5): $K = -\frac{1}{3} \,\mathrm{Tr}\left(e^{gV} \bar\Phi_I e^{-gV} \Phi^I\right)$, the K of (7)–(8), whereas (1) uses $K = \mathrm{Tr}\, \bar\Phi e^{gV} \Phi$ without $-\frac{1}{3}$; footnote 8: $\bar D^2 = -\frac{1}{4} \bar D_{\dot\alpha} \bar D^{\dot\alpha}$; $Z_B = 1$, $Z_F = \frac{g^2 N}{32\pi^2}$ at order g^2 in the text after (12), with C_F of (10); SSDR; [BKRS01]: g of their refs $= 2g$ (footnote 3), $W_\alpha = -\frac{1}{8g} \bar D^2 (e^{-2gV} D_\alpha e^{2gV})$ (8).

(c) [I] Untwisted operator (contracted, $|O| = 0$, $w_O = \frac{5}{2}$):

$$O = \int \omega \left(u'^\alpha \lambda_\alpha{}^i,\ A_{ij} w^j\right),$$

index-free: $\int \omega$ of the $(-,-)$ pairing of the u-coefficient of $\lambda|_{S_+ \otimes W}$ with the contraction of $A|_{W^\perp_{\mathbb{R}}}$ with w. Classical component along ξ: by $\xi.(u'\lambda^i) = \frac{1}{4}[A^{ik}, A_{kj} w^j]$ modulo w^i and $\xi.(A_{ij} w^j) = 0$, it is $\pm\frac{1}{4} \int \omega\, ([A^{ik}, A_{kj} w^j], A_{il} w^l)$ (sign fixed by the bracket conventions in code), the image of $3W[\gamma]$ ([`gkw.tex`] inference I5: $Q_0(\sum_I \mathrm{Tr}\, \beta_I \gamma^I) \ni 3W[\gamma]$). One-loop target: arity one ($k = 2(3-\frac{5}{2}) = 1 = 1 + 0 - 0$), order $\hbar$, output

$$c_K \,\frac{\hbar \,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X \mathrm{ad}_Y)}{16\pi^2 (X,Y)} \int \omega \left(\bar\lambda_{\dot\alpha i} w^i,\ \bar\lambda^{\dot\alpha}{}_j w^j\right),$$

the image of $\mathrm{Tr}\, \bar\Psi^4_{\dot\alpha} \bar\Psi^{4\dot\alpha}$. From [BK] (3.66): $\sum_I Q_1(\delta_{AB}(\beta_I)^A(\gamma^I)^B) = \frac{3}{2} f_{ACD} f_{ACE} \partial c^D \cdot \partial c^E = -\frac{3}{2} \,\mathrm{tr}(\mathrm{ad}\, t_D \,\mathrm{ad}\, t_E)\, \partial c^D \cdot \partial c^E$ [D]; the number c_K follows only after the normalised field map and the relation of the twisted $\hbar$ to the project's unit are derived (open item 1). Physical cross-checks, as statements: [EJSS05] (8), [BKRS01] (9), [CL25] (2.5). The identification of the [EJSS05] coefficient with $-1\times$unit is withdrawn: the dictionary (Cor. cor:wick) uses $\mathrm{tr}(T^a T^b) = \frac{1}{2}\delta^{ab}$ with $(X,Y) = -\frac{4\hbar}{g^2} \,\mathrm{tr}(XY)$, [EJSS05] uses

$\mathrm{Tr}(T^aT^b)=\delta^{ab}$, e^{gV}, $K=-\frac{1}{3}\,\mathrm{Tr}$ (5) and $\bar D^2=-\frac14\bar D\bar D$ (footnote 8), and the [EJSS05] number is a coefficient in a superfield operator relation, not an arity-one component.

(d) [D] Scheme dependent: under an operator redefinition φ_1 with $\varphi_1(W[\gamma])=x\,\mathrm{Tr}\,\partial c\cdot\partial c$ (allowed: [BKRS01] (9)–(10) and [EJSS05] (17), (99) show that the trilinear B and the bilinear F have the same quantum numbers and mix), $Q_1\mapsto Q_1+[Q_0,\varphi_1]$ changes the coefficient of $\mathrm{Tr}\,\partial c\cdot\partial c$ in $Q_1(\sum_I \mathrm{Tr}\,\beta_I\gamma^I)$ by $-3x$ (from $Q_0(\sum_I \mathrm{Tr}\,\beta_I\gamma^I)\ni 3W$), modulo Q_0-exact terms. This is the twisted form of [EJSS05] (2): $a(g),b(g)$ "depend on the normalization of the operators, i.e. on the subtraction scheme"; the [EJSS05] abstract states more: in a superconformal renormalisation scheme "the separation into 'classical' and 'quantum' anomaly terms is not possible", and (3) contains no $[F]_R$ term. [D] The coefficient c_K remains scheme dependent after two-point orthogonality: [EJSS05] (101)–(106) construct $K_{10}=Z_KB+Z_OD^2O$ orthogonal to D^2O ((103)) with Z_K of minimal-subtraction form by choice ((102)) and $-2D^2O=F-4B$ ((99)); the redefinition $B'=B+yg^2(F-4B)$ has the same quantum numbers (the project's F3 freedom), keeps (99) up to normalisation, and orthogonality then gives the F coefficient $\frac{g^2N}{32\pi^2}-yg^2$. [EJSS05] footnote 13: "This may be a scheme-dependent property." In the twisted language $\sum_I \mathrm{Tr}\,\beta_I\gamma^I$ is not Q_0-closed, so c_K is not part of $\pi_0Q_1\iota_0$ and is scheme dependent by Lemma 3. In the SO(7) lifting of [CL25] §3, Q_K is applied to the Q_0-closed O_f ($Q_KO_f=O_m+Q_0\Lambda$); the classical term never enters, only the nonvanishing of the overall normalisation of Q_K (Target 8). [CDSW] after (3.4): the higher-loop contributions to the Konishi anomaly "are renormalization scheme dependent" and are non-chiral local functionals, so the chiral-ring relation (3.4) is exact.

Target 6 (the remaining $\mathcal N=4$ pair entries)**.** (a) [S] [BK] (3.65): $Q_1(b^Ab^B)=\frac12 f_{ACD}f_{BCE}[\partial c^D\cdot\partial b^E-\partial b^D\cdot\partial c^E+\partial(\beta_I)^D\cdot\partial(\gamma^I)^E-\partial(\gamma^I)^D\cdot\partial(\beta_I)^E]$; (3.67): $Q_1((\beta_I)^A(\beta_J)^B)=\frac12\varepsilon_{IJK}f_{ACD}f_{BCE}[\partial c^D\cdot\partial(\gamma^K)^E-\partial(\gamma^K)^D\cdot\partial c^E]$; (3.68): $Q_1(b^A(\gamma^I)^B)=\frac12 f_{ACD}f_{BCE}[\partial c^D\cdot\partial(\gamma^I)^E-\partial(\gamma^I)^D\cdot\partial c^E]$; (3.69): $Q_1((\beta_I)^Ab^B)=\frac12 f_{ACD}f_{BCE}[\partial(\beta_I)^D\cdot\partial c^E+\partial c^D\cdot\partial(\beta_I)^E+\varepsilon_{IJK}\partial(\gamma^J)^D\cdot\partial(\gamma^K)^E]$. Superfield (3.70)=(1.10): $Q_1(C^A(\theta)C^B(\theta'))=-\frac12 f_{ACD}f_{BCE}(\theta_1-\theta_1')(\theta_2-\theta_2')(\theta_3-\theta_3')\,\partial_{\dot\alpha}C^D(\theta)\partial^{\dot\alpha}C^E(\theta')$, $C=c+\theta_I\gamma^I+\frac12\varepsilon^{IJK}\theta_I\theta_J\beta_K+\theta_1\theta_2\theta_3 b$ ((3.48)–(3.54)). Derivation from $Q_1(bc)$ by tree-level $\mathfrak{psl}(3|3)$: (3.59)–(3.63), e.g. $Q_1(b\gamma^I)=-Q_+^IQ_1(bc)$, $Q_1(\beta_I\gamma^J)=-\frac12 S_I^+Q_1(b\gamma^J)$. Derivatives: (B.1) $Q_1(\partial_{\dot\alpha}f\,g)=\partial_{\dot\alpha}Q_1(fg)-Q_1(f\partial_{\dot\alpha}g)$, kernel (B.4), (B.6)–(B.10). Chang–Yin-letter versions: [KKLP] (2.5)–(2.12).

(b) As Target 4; (3.70) not re-derived by the readers.

(c) [D] Arity assignments (Lemma 1) with the field representatives of the letter table; every entry is consistent:

entry	$(w_{\rm in},g_{\rm in})$	$(w_{\rm out},g_{\rm out})$	k
bc (3.64)	$(2,1)$	$(3,0)$	2
bb (3.65)	$(4,0)$	$(\frac92,0)$	1
$\beta\gamma$ (3.66)	$(\frac52,0)$	$(3,0)$	1
$\beta\beta$ (3.67)	$(3,0)$	$(\frac72,0)$	1
$b\gamma$ (3.68)	$(3,0)$	$(\frac72,0)$	1
βb (3.69)	$(\frac72,0)$	$(4,0)$	1

Untwisted operators: replace $b,c,\gamma,\beta,\partial_{\dot\alpha}c,\partial_{\dot\alpha}$ by the entries of the letter table, e.g. $(\beta_I)^A(\beta_J)^B\leftrightarrow(X,u'\lambda^i)(Y,u'\lambda^j)$ and $\partial c\cdot\partial\gamma\leftrightarrow(\bar\lambda_{\dot\alpha i}w^i)\,(u'^\alpha\sigma_\alpha^{m\dot\alpha}D_m(A_{kj}w^j))$.

(d) Pair values scheme dependent; the relations (3.59)–(3.63), as relations between classes on $H(Q_0)$, follow from $\{Q_1,R_0\}+\{Q_0,R_1\}=0$ (Target 10) and are scheme independent at the class level.

Target 7 (no Leibniz rule)**.** (a) [S] [BK] footnote 21: generically no associative scheme in which $\mathcal Q$ obeys the Leibniz rule. [BGKWWY] footnote 28, (3.26): Leibniz holds only for the loop-corrected product, which is not associative; (3.26) fails by a one-loop term reproducing the anomaly. [CL22] (3.4)–(3.5): if $\mathcal Q$ obeys Leibniz, $H^*(\mathcal G_N,\mathfrak{sl}_N;\mathbb C[[g_{YM}]])\cong H^*(\mathcal G_N,\mathfrak{sl}_N;\mathbb C)\otimes\mathbb C[[g_{YM}]]$; main text p. 9: "the only known obstruction, the Konishi anomaly [63], is absent in $\mathcal N=4$ SYM [64]"; footnote 10: a scheme with undeformed $\mathcal Q$ on the BPS superfield via DRED [61] or the scheme of [62]; footnote 11 is

an acknowledgement (D. Gaiotto, J. Kulp). [I] The statement that the Konishi anomaly is absent in $\mathcal{N}=4$ is in tension with the nonzero Konishi-type pair entries of [BK] (3.66) and [CL25] (2.5).

(c) [I] The arity-one component along ξ at order $\hbar$ is not a derivation of the classical product of letters modulo Q_0-exact terms; Targets 5 and 8 give the explicit obstruction.

(d) [I] Scheme independent: the existence of $[Q_1]$-lifted pairs (Target 8) excludes $Q_1 = aQ_0 + [Q_0, \varphi]$ (Lemma 3), and excludes a scheme in which both hypotheses of [CL22] p. 9 hold, (i) $\mathcal{Q}$ undeformed on the BPS superfield and (ii) the Leibniz rule. A nonzero derivation Q_1 is not excluded by this argument: a derivation that (anti)commutes with Q_0 can act nontrivially on $H(Q_0)$, e.g. [BK] (3.55) $Q_+^I \sim \partial_{\theta_I}$. The DRED scheme of [CL22] footnote 10 is excluded in this project (no dimensional regularisation in any role).

Target 8 (lifting statements on $H(Q_0)$)**.** (a) [S] [CL25] §3: in SO(7), $Q_K O_f = O_m + Q_0\Lambda$, $O_m \propto M_1 - M_2 - 4M_3$ independent of the 741 Q_0-exact operators of the sector ($J_a = 0$, $R_i = 3$, 8 letters), $M_1 + 2M_2 - 10M_3$ Q_0-exact (3.3); resolves the SO(7)/Sp(3) mismatch of [arXiv:2510.24008]. [KvLM] (1.2)/(5.3): $[Q_1][F_4] = c[S^4]$, $c \neq 0$, $S = \mathrm{Tr}(\lambda_+\lambda_-)$, SO(7); §6: $[Q_1]$ annihilates every monotone class; every nonzero $[Q_1]$-image has vanishing Cartan restriction; key identity (1.3) $\Delta^k F = Q_0 H$ for Q_0-closed F with vanishing Cartan restriction. [KKLP] abstract, §3: $SU(2)$, the lightest fortuitous class and its hairy versions are not lifted, many heavier core fortuitous classes are lifted. [KLLS] abstract, §2–3: SU(3) brane-fused family not lifted at first quantum order; several black-hole cohomologies lifted; normalisation of Q_1 plays no role (after (2.5)). [BK] §3.4.1 (see Target 3). [BMV] (abstract, main text): the one-loop dilatation operator of $\mathcal{N}=4$ SYM at non-integer N; the lightest classically $\frac{1}{16}$-BPS black-hole state has protected dimension $\Delta = \frac{19}{2}$ at $N=2$ and becomes, at infinite N, a triple trace of Konishi and two BPS operators with $\Delta = \frac{19}{2} + 12\lambda + \cdots$; [BK] p. 5 credits "directly computing the one-loop action [37]" (=[BMV]) for the survival of the lowest $SU(2)$ fortuitous class, an untwisted one-loop computation; [BMV] footnote 5: the first higher-loop correction of [BGKWWY] preserves the $SU(2)$ threshold class O_0.

(b) Letters of [CY13] (2.2)–(2.3) ($\phi^n = \Phi^{4n}$, $\psi_n = -i\Psi_{n+}$, $\lambda_{\dot\alpha} = \bar\Psi^4_{\dot\alpha}$, $f = -iF_{++}$, $D_{\dot\alpha} = D_{+\dot\alpha}$); tree cohomology $H^*(\mathcal{G}_N, \mathfrak{sl}_N; \mathbb{C})$ ([CY13] (3.6)).

(c) [I] Statements about $\pi_0 Q_1 \iota_0$ on the classical 1/16-BPS cohomology, i.e. about the order-$\hbar$ part of the transported $\mathrm{ev}_\xi \circ (d_{CE} + Q)$ on $H(Q_0)$. In the untwisted program they follow from Targets 2–6 once the pair entries are reproduced up to the terms of Section 4; a direct untwisted computation on these multi-letter classes is outside Steps 2–3.

(d) Scheme independent ([BK] p. 5 and footnote 9: the lifting of cohomology classes is scheme independent, assuming non-perturbative effects only reduce the cohomology; Lemma 3).

Target 9 (stress tensor and the absent one-loop beta function)**.** (a) [S] [BK] (2.17)–(2.18): $S^{(0)}_{\dot\alpha} := S_{+\dot\alpha,+}$, $D_- S_{\dot\alpha} = (\mathcal{Q} + \bar\partial) S_{\dot\alpha} = 0$ for superconformal theories. Contrast, pure $\mathcal{N}=1$: [BGKWWY] (3.40) $Q_1 \,\mathrm{Tr}\, b^2 \propto \partial_\alpha(\mathrm{Tr}\, b\partial^\alpha c)$ (holomorphic confinement), (3.24) $(Q+\bar\partial)S_\alpha \sim \partial_\alpha \,\mathrm{Tr}(\partial_\beta c \partial^\beta c)$, (2.20)–(2.22): for a classically scale-invariant theory with a one-loop beta function $(Q+\bar\partial)S$ is nonzero at one loop. Reader checks (`bgkwwy_checks`, CK02, CK05; not source statements): for $SU(2)$, $Q_1 \,\mathrm{Tr}\, b^2 = -4\,\partial_\alpha \,\mathrm{Tr}(b\partial^\alpha c)$ from (C.3), in the [BGKWWY] normalisation (twice the [BK] one, Target 4), and the coefficient in (3.24) is 1 with (C.6) normalised to (C.4).

(c) [I] Untwisted: the coupling is not renormalised at one loop in the project (Step 1: coefficient along the coupling direction zero, STORY.md); the twisted image is that $[Q_1]$ annihilates the class of the holomorphic stress tensor. [D] There is no $\mathcal{N}=4$ analogue of the [BGKWWY] (3.40) lifting: by [BK] (3.49), $Q_0 b = [c, b] - [\gamma^I, \beta_I]$, so $Q_0 \,\mathrm{Tr}\, b^2 = -2\,\mathrm{Tr}\, b[\gamma^I, \beta_I] \neq 0$ and $\mathrm{Tr}\, b^2$ is not Q_0-closed; from [BK] (3.65), $Q_1 \,\mathrm{Tr}\, b^2$ is proportional to $\partial^{\dot\alpha}\big(\mathrm{Tr}\, b\,\partial_{\dot\alpha} c + \mathrm{Tr}\,\gamma^I \partial_{\dot\alpha}\beta_I\big)$, a total derivative of the stress-tensor type of [BGWW] (4.82); this is a check of compatibility with [BK] (2.18). The untwisted operator is the component of the stress-tensor multiplet selected by (u, u', w) ([BK] (2.17)); its two-component form is a Step-3/4 item. The pure-$\mathcal{N}=1$ statements are not reproducible inside this $\mathcal{N}=4$ program and serve as contrast only.

(d) Scheme independent (class level).

Target 10 (compatibility with the tree-level $\mathfrak{psl}(3|3)$)**.** (a) [S] [BK] (3.55)–(3.58): tree-level $\mathfrak{psl}(3|3)$ commuting with Q_-^4: $Q_+^I \sim \partial_{\theta_I}$, $S_I^+ \sim \theta_I(z^{\dot\alpha}\partial_{\dot\alpha} + \theta_J\partial_{\theta_J})$, $\bar Q_{I\dot\alpha} \sim \theta_I\partial_{\dot\alpha}$, $\bar S^{I\dot\alpha} \sim z^{\dot\alpha}\partial_{\theta_I}$; footnote 24: "The tree-level action of $\mathfrak{psl}(3|3)$ might also receive loop corrections, however the one-loop differential Q_1 (anti)commutes with the tree-level generators R_0 ... i.e. $\{Q_1, R_0\} + \{Q_0, R_1\} = 0$"; [I] R_1 is not written out in [BK]. [KvLM] §5, paragraph after (5.4): the classical commutant supercharges $\bar Q_{m\dot\alpha}$ anticommute with Q_1 in Q_0-cohomology, cited from [12]=[BK]; (5.4) itself is the classical action of six $\bar S$ commutant charges on $\mathcal{S}^4$, proportional to $M_1 - 3M_2$.

(c) [D] Elements of $\mathfrak{psl}(4|4)$ with vanishing bracket with $\xi = u \otimes w$ (Def. def:c2-bracket, (B1)–(B2) of the dictionary): all of $S_+ \otimes W$; $\bar\xi' \in S_- \otimes W^*$ with $\bar\xi'_{\dot\alpha i} w^i = 0$; $\eta \in S_+ \otimes W^*$ of the form $\eta^\alpha{}_i = u^\alpha m_i$ with $m_i w^i = 0$; all of $S_- \otimes W$. [BK]'s generators are among these (labels as printed). [I] The BV form of footnote 24 is the order-$\hbar$ part of the (ξ, R) component of $(d_{CE} + Q)^2 O[\infty] = 0$; R_1 is the arity-one component along R at order $\hbar$, for $R \in \Sigma^*$ a special-supercharge component S_1 of directive R4.

(d) The class-level relation is scheme independent; R_1 is scheme dependent jointly with Q_1 (compare [`n4_untwisted` D2]).

Target 11 (scheme statements to be reproduced as theorems)**.** (a) [S] [BGKWWY] §3.2.1: **Q** scheme dependent, schemes related by quasi-isomorphisms; App. E: parametrisation of deformations scheme dependent beyond first order. [BK] §2.2.2: Q_n scheme dependent beyond first order, a scheme may exist where they vanish at some order, schemes related by quasi-isomorphisms of L_∞-conformal algebras; p. 5, footnote 9: lifting scheme independent. [GKW] §2 after (2.10), after (2.17), footnote 13: η scheme dependent, schemes related by L_∞ quasi-isomorphisms; translations not corrected (footnote 13, (2.31)); [GKW] (6.28) (BRST anomaly of the regularised interaction) and (6.33) (brackets as contour brackets) are not scheme statements. [CDSW] after (3.4): higher-loop contributions to the Konishi anomaly are "renormalization scheme dependent". [BGKWY22] App. B (B.6): a change $P \to P + d_y H$ of the propagator changes I_Γ by boundary terms, "essentially an infinitesimal operator redefinition". [W18] Thm. 3.4: holomorphic theories on $\mathbb{C}^d$ admit a one-loop prequantisation without counterterms. Spin chains: [B03] (3.39)–(3.40), (4.4)–(4.10): similarity $J \mapsto TJT^{-1}$, $J_2 \mapsto J_2 + [T_2, J_0]$, coupling $g \mapsto f(g)$; [Z05] (3.7), (4.1)–(4.2). Derived [`n4_untwisted` D2]: in su(2|3), $Q_2 = 0$ iff $\gamma = \frac{i\alpha_1^2}{4}(1, 1, -1, 1)$, then $S_2 \neq 0$; $Q_2 = S_2 = 0$ is impossible; $\{S, Q\}_2 = \frac{1}{2}\delta\delta H_2$ is invariant.

(c) Project counterpart: $O(\varepsilon)$ has a unique singular part and a free finite part (DEFINITIONS_v2 item 5); finite counterterms in $C^\bullet(\mathfrak{psl}(4|4), \mathscr{O}_l(\mathscr{E}))$ (ch07, Step 2). Section 4.

Target 12 (the rank-$(1,1)$ supercharge: no loop corrections)**.** (a) [S] [CLW] (3.1): the holomorphic-topological supercharge $Q_{HT} = Q_-^4 + Q_{\dot -3}$. [CLW] §3.1: "all the loop diagrams naively contributing to the BRST differential vanish identically" (the $(H+T)$-Laman graphs with $H = 1$, $T = 2$, via the vanishing of the top forms $\omega_\Gamma^{H,T}$ on 3-Laman graphs), so $Q_{\rm int}$ truncates at tree level to Q_0 and obeys the Leibniz rule, to all orders. [CLW] footnote 5: the leading loop correction Q_1 computed in [32]=[Z05] "takes the form $[Q_0, -]$". [CLW] (3.17)–(3.19), [BK] footnote 6, (1.11): in the 2d chiral algebra the one-loop correction Q_1^{2d} is nonzero, $Q_1^{2d}(C^A(\theta)C^B(\theta')) = -2f^{ABC}(\theta^1 - \theta'^1)(\theta^2 - \theta'^2)\partial C^C$ ((3.19)).

(c) [D] In the project Q_{HT} is $x = \xi + \bar\xi'$ with $\xi = u \otimes w$ and $\bar\xi' \in S_- \otimes W^*$ of rank one with $\bar\xi'_{\dot\alpha i} w^i = 0$ (the commutant of Target 10 (c)); $\Gamma(x, x) = 0$ in all ten components, and the evaluation along x and the twisted action of the design (lem:o-ev) apply. Target: $[Q_1^x] = 0$ on $H(Q_0^x)$, from the components along ξ, $\bar\xi'$ and $(\xi, \bar\xi')$.

(d) Scheme independent: the class statement $[Q_1^x] = 0$ (Lemma 3); the pair values vanish only in the HT scheme of [CLW]. This addresses open item 6 (the [Z05] tension) and is an R3 reproduction check.

4 Renormalisation freedom of Q_1 (R3 display)

Let $\mathcal{Q}^{\rm tw} = Q_0 + \hbar Q_1 + O(\hbar^2)$ be the twisted differential on twisted letters obtained from $\mathrm{ev}_\xi \circ (d_{CE} + Q)$ and transport. The untwisted choices act as follows (each item [I] unless marked; all to be established in Step 2/3).

F0. Arity-zero finite counterterms $F^{(0)}_{\mathfrak{psl}(4|4)}$ (DESIGN_STEP2 Part C; design prop:o-arity0): two-

leg part $\frac{27}{16}\,\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}\int(d^*A|_V,d^*A|_V)$ (forced) $+\beta_1\int(dA|_V,dA|_V)+\beta_2\int(dA|_{W^\perp},dA|_{W^\perp})+\beta_3\int(\lambda,\not{\partial}\lambda)+\beta_4\int(dc,A^*)$, three-leg part in $\mathrm{span}\{H_1,\dots,H_6,H_7,H_{8+},H_{8-}\}$ with $H_7=\frac12\int(\lambda,\rho([A|_V,\lambda]))$ and $H_{8+}+H_{8-}=\frac12\int(\lambda,\rho([A|_{W^\perp},\lambda]))$ split by chirality ($QH_7=-G_5$, $QH_{8\pm}=0$, cubic_structure prop:s2c-QH and the proof of cor:s2c-QI3). [D] The closed cubic-order directions form a space of dimension 9 (8 parity-even); the five field-counting $s\phi$ and the coupling direction span 6 of them (5 parity-even), the Part-C family (three-leg part in $\mathrm{span}\{H_1,\dots,H_6\}$) has dimension 6 and meets that span in dimension 3, and the quotient has dimension 3, all parity-even, with representatives (β_1,H_1), $(\frac14\beta_2,H_4)$, $(-\frac12\beta_3,H_5)$; the Yukawa direction $H_{8+}+H_{8-}$ and H_5+H_7 are closed and lie in that quotient. Two-component: $H_7=\int i(\bar\lambda_{\dot\alpha i},\bar\sigma^{m\dot\alpha\alpha}[A_m,\lambda_\alpha{}^i])$, $H_{8+}=\frac12\int(\lambda_\alpha{}^j,\varepsilon^{\alpha\beta}[A_{ji},\lambda_\beta{}^i])$, $H_{8-}=-\frac12\int(\bar\lambda_{\dot\alpha j},\varepsilon^{\dot\alpha\dot\beta}[A^{ji},\bar\lambda_{\dot\beta i}])$. Two-component: $\beta_1\int(dA|_V,dA|_V)=\frac{\beta_1}{2}\int(F^0_{mn},F^0_{mn})$, $\beta_2\int(dA|_{W^\perp},dA|_{W^\perp})=\frac{\beta_2}{4}\int(\partial_mA^{ij},\partial_mA_{ij})$, $\beta_3\int(\lambda,\not{\partial}\lambda)=2i\beta_3\int(\bar\lambda_{\dot\alpha i},\bar\sigma^{m\dot\alpha\alpha}\partial_m\lambda_\alpha{}^i)$, $\beta_4\int(\partial_mc,A^*_m)$ (dictionary, Def. def:bases). The β-directions complete to s-closed functionals (DESIGN_STEP2, "solution family"): their s-exact part is a field-type rescaling N_β, acting as $Q_1\mapsto Q_1+[Q_0,N_\beta]$; the coupling direction (total action, $G=S-S^{(2)}$, design prop:o-laws (d), $F\mapsto F+\hbar aG$) acts as $Q_1\mapsto Q_1-aQ_0-a[N_{R_\xi},Q_0]$: the aQ_0 part is the overall rescaling of the twisted action, the second part the R-rotation of F1; untwisted, it changes the one-loop BRST transformation by $-a(Q+\{I,-\})$, which vanishes on gauge-invariant operators.

F1. Arity-one finite counterterm $F(\xi)$ (ch07 "The road ahead": obstruction $\mathrm{Ob}(\xi)=(Q+\{I,-\})F(\xi)+\{S^{(1)}(\xi),F^{(0)}_{\mathfrak{psl}(4|4)}\}$): unique up to s-closed local functionals depending linearly and equivariantly on $\xi\in S_+\otimes W$; by Schur's lemma for the covariant subalgebra the closed choice along ξ is $a\,S^{(1)}(\xi)$, a renormalisation of the supercharge. With the correlated F2 rescaling, $S+(1+a\hbar)S^{(1)}(\xi)-\frac12(1+2a\hbar)S^{(2)}(\xi,\xi)$ is S^ξ composed with the R-rotation $\alpha(1+a\hbar)^{-1}$ of [ESW, Def 2.18] at first order (S and ω are α-invariant, $S^{(k)}(\xi,\dots,\xi)$ has α-weight k), so F1 acts by conjugation: $Q_1\mapsto Q_1+a[N_\alpha,Q_0]=Q_1+aQ_0-a[N_{\mathrm{gh}},Q_0]$, a $[Q_0,\varphi]$-type shift, zero on c and γ and the term $\frac12\varepsilon_{IJK}[\gamma^J,\gamma^K]$ on β_I up to the sign of the letter map (design rem:o-alpha). In the untwisted equivariant setting $a\,S^{(1)}(\xi)$ is the restriction to ξ both of the outer cocycle G_B (F5) and of the exact cocycle of the inner R-rotation R_ξ; s-exact choices act as $[Q_0,-]$. The parameter a of F1 (added to the action) is minus the a of design prop:o-laws (e) ($F\mapsto F+\hbar aG_B$ changes the action by $-a\hbar G_B$, and $r_\xi(G_B)=S^{(1)}(\xi)-S^{(2)}(\xi,\xi)$).

F2. Arity-two finite counterterm $F(\xi,\xi)$: correlated with F1 by the (ξ,ξ) component of the master equation; [D] the equation $\{S,S^{(2)}(\xi,\xi)\}+(-1)^{|\xi|}\{S^{(1)}(\xi),S^{(1)}(\xi)\}=0$ (lem:c2-components, $[\xi,\xi]=0$) is linear in $S^{(2)}$ and quadratic in $S^{(1)}$, so $S^{(1)}(\xi)\mapsto(1+a\hbar)S^{(1)}(\xi)$ forces $S^{(2)}(\xi,\xi)\mapsto(1+2a\hbar)S^{(2)}(\xi,\xi)+O(\hbar^2)$; by Lemma 2 the change affects only outputs with λ^*-legs at one loop.

F3. Finite parts of operators: $O(\varepsilon)\mapsto O(\varepsilon)+\hbar\varphi_1(O)$, $\varphi_1(O)$ local with the Weyl weight, ghost degree and $\mathfrak{so}(V)\oplus\mathfrak{so}(W^\perp_{\mathbb{R}})$ representation of O (e.g. the B–F mixing of Target 5): $Q_1\mapsto Q_1+[Q_0,\varphi_1]$.

F4. Choice of representatives and homotopy data in the transport ([BGKWWY] (D.14) h_0; [ESW] Thm. 10.9 eliminations; [BGKWY22] (B.6) propagator changes): $Q_1\mapsto Q_1+[Q_0,\varphi_h]$.

F5. [D] The outer derivation B of $\mathfrak{psl}(4|4)$ (induced by the bracket with $\mathrm{diag}(0_4,1_4)$ in $\mathfrak{gl}(4|4)$, equivalently minus the bracket with $\mathrm{diag}(1_4,0_4)$, ch08 Def c8-B; $+1$ on $S_+\otimes W$ and $S_-\otimes W$, -1 on $S_-\otimes W^*$ and $S_+\otimes W^*$, 0 on the even part): $G_B=\frac{d}{dt}(e^{tB})^*S_{\mathfrak{psl}(4|4)}$ is s-closed and not s-exact (design prop:o-GB). Untwisted ($F\mapsto F+\hbar aG_B$): $Q_1\mapsto Q_1-a\hbar\,\mathrm{ad}_{S^{(1)}(\xi)}$ on $S_+\otimes W$, $+a\hbar\,\mathrm{ad}_{S^{(1)}(\xi)}$ on $S_-\otimes W^*$; $S_1\mapsto S_1+a\hbar\,\mathrm{ad}_{S^{(1)}(\eta)}$ on $S_+\otimes W^*$, $-a\hbar\,\mathrm{ad}_{S^{(1)}(\eta)}$ on $S_-\otimes W$; D_1,K_1,P_1,M_1,R_1 unchanged. Twisted: $Q_1\mapsto Q_1+a[N_{R_\xi},Q_0]$, of the form of Lemma 3.

Lemma 3 (invariance, [D])**.** If $Q_1'=Q_1+aQ_0+[Q_0,\varphi]$ with $Q_0'=Q_0$, then $\pi_0Q_1'\iota_0=\pi_0Q_1\iota_0$ on $H(Q_0)$, because $\pi_0Q_0=0$ and $Q_0\iota_0=0$. Hence the class map $[Q_1]$ (the leading term $\hbar Q_0^{(1)}$ of [BGKWWY] (D.15)), its kernel, image and rank, and the lifting pattern are independent of F0–F4. Rescalings of the letters at tree level conjugate $[Q_1]$ by an invertible map of $H(Q_0)$ and leave kernel, image and rank unchanged. The shifts of F0, F1 and F5, $-aQ_0-a[N_{R_\xi},Q_0]$, $a[N_\alpha,Q_0]$ and $a[N_{R_\xi},Q_0]$, are of this form, since $[N,Q_0]=-[Q_0,N]$ for an even derivation N.

To be displayed as families (scheme dependent): every pair value (3.64)–(3.69) and their shifted/derivative versions; the vanishing on single letters and the pairwise form (Target 2); "triangle only"; the coefficient c_K of the Konishi entry, in every normalisation of the weight-3 operators, orthogonality to the protected operator included (Target 5); the overall size of Q_1 relative to Q_0 up to tree-level letter rescalings.

To be reproduced exactly (scheme independent): $\{Q_0, Q_1\} = 0$ and $[Q_1]^2 = 0$ on $H(Q_0)$; $[Q_1]$ on $H(Q_0)$, in particular the class-level statements of Targets 3, 8, 9, 10; the exact vanishing of the component along ξ on $\int \omega\,(\iota_{w_1} A, \iota_{w_2} A)$ with w_1, w_2 in the plane of w (Target 3, a selection-rule consequence); $[Q_1^x] = 0$ for the rank-$(1,1)$ supercharge (Target 12).

5 What the literature does not have: S, K, dilatation at one loop (R4)

generator	known (source)	not found (evidence)
$\xi \in S_+ \otimes W$ (twisting Q)	twisted Q_1 complete: [BK] (3.64)–(3.70), App. B; untwisted classical Q on letters [CY13] (2.5)–(2.8), [GGKM08] (A.1)–(A.2); Konishi entry only [CL25] (2.5)	untwisted one-loop Q along Q^4_- on the full letter set, including two-to-one terms with derivatives (`n4_untwisted.json` open issues); Q_1 as a component of an untwisted equivariant BV cochain (`sweep.json` cross-paper "not found")
special supercharges $\eta \in \Sigma^*$ (S_1)	planar full chain: [Z11] §2.3, (2.33): the $2 \to 1$ parts $(S^C_\alpha)_{2\to1}$, $(\dot S^C_{\dot\alpha})_{2\to1}$ through the generalised harmonic action $H^{(1/2,1/2)}$, derived in Zwiebel's thesis, "fixed (up to normalization) by superconformal symmetry", one loop by [`n4_untwisted` D1] (order g, $L = (1-1+2)/2$); planar su(2|3): [B03] (3.37), Table 1 (S_1 at $O(g)$, S_2 at $O(g^2)$ with $\gamma_{1..4}$), S_3 only counted (Table 2, §5); psu(1,1|2): [Z05] (3.6) $\mathfrak{X}^-_2 = -[\mathfrak{X}^-_0, \mathfrak{x}] + [\mathfrak{X}^-_0, \mathfrak{y}]$; conjecture [B03c] (5.15) $\delta S(g) \sim g \operatorname{Tr} \Psi \mathcal{Q}$	the $2 \to 2$ part of the one-loop S outside su(2|3) and psu(1,1|2); the non-planar part; gauge-variant letters; the BV-equivariant form; twisted R_1 for S^+_I, $\bar S^{I\dot\alpha}$ ([I] not written out in [BK], footnote 24); S as BV-equivariant component (reader file `bgkwwy.json`, entry `absent_from_paper`; `gkw.json` I6)
special conformal K (K_1)	planar full chain: [Z11] (2.34): $(K_{\alpha\dot\beta})_{2\to1} = \{(S^1_\alpha)_{2\to1}, (\dot S_{1\dot\beta})_{1\to1}\} + \{(S^1_\alpha)_{1\to1}, (\dot S_{1\dot\beta})_{2\to1}\}$ (one loop by D1) and $(K_{\alpha\dot\beta})_{3\to1} = \{(S^1_\alpha)_{2\to1}, (\dot S_{1\dot\beta})_{2\to1}\}$ (two loops by D1); psu(1,1|2): [Z05] (3.6) $\mathfrak{K}^{22}$ at $O(g^2)$, homogeneous solutions neither found nor excluded ((3.8)–(3.9)); [B03c] §6: K corrected, not computed; derived [`n4_untwisted` D4]: with P uncorrected ($\mathfrak{y} = -\mathfrak{x}$), $\mathfrak{K}^{22}_2 = -2[\mathfrak{K}^{22}_0, \mathfrak{x}] \neq 0$	the $2 \to 2$ part of the one-loop K outside psu(1,1|2); the non-planar part; gauge-variant letters; the BV-equivariant form; any twisted-side K (the twisted theory keeps only holomorphic transformations; [BGWW] (3.47)–(3.52) treat anomalies of holomorphic symmetries as failures of an L-equivariant QME and list only classical $\mathcal{N} = 4$ currents (4.82)–(4.85))

dilatation Δ (D_1)	complete one-loop D_2 on all local operators [B03c] (2.22), (2.24), (4.27)–(4.29) (point splitting, (3.14)); Konishi [B03c] (4.49): $E = 6$, $\delta\Delta = \frac{3g_{YM}^2 N}{4\pi^2}$ with the normalisation (1.2) $D(g) = D_0 + \frac{g_{YM}^2 N}{8\pi^2} H + O(g^3)$, $= -24\times$unit (`n4_untwisted` D7, ch06 rem:c6-konishi)	D_1 as the Δ-component of an equivariant BV cochain with a UV-boundary regulator, and its identification with the scaling route (DESIGN_STEP2 Part D, D3); gauge-variant letters in a stated gauge
P, M, R	expected uncorrected: [GKW] footnote 13, (2.31); [B04] (2.31): $L, \dot{L}, R$ not corrected; [B03c] §6 lists P among corrected generators in its basis (similarity dependent); project Prop. prop:c7-cov: can be chosen classical	realisation by an explicit equivariant choice of counterterms (Step 2)
arity two (ξ,η), (ξ,ξ)	su(2|3): [B03] (3.41)–(3.42), (4.3) $\{S,Q\} = \cdots + \delta_a^b \delta_\beta^\alpha (\frac{1}{3} H + \frac{1}{6}\delta H) + O(g^4)$; [`n4_untwisted` D2] invariance of $\{S,Q\}_2$	BV form $((d_{CE} + Q)^2 O[\infty] = 0$ on (ξ,η), cf. rem:c7-KDelta for $(K,P))$: not found
equivariant one-loop QME along all $\mathfrak{psl}(4\|4)$ ghosts	twisted analogue for holomorphic symmetries [BGWW] (3.46)–(3.49); nearest twisted one-loop data along superconformal generators: [BGWW] §4.3, (4.73)–(4.77), the ternary λ-brackets of the twisted $\mathcal{N} = 4$ superconformal currents, encoding $a = c$ (c-number data), with the representative of the 3-cocycle "for future work"; [CLW] §3.1: no loop corrections along Q_{HT}	untwisted: not found (`sweep.json` "not found in the sweep")

Consequence for R4 [I]: the outer derivation B (F5) shifts Q_1 and S_1 by opposite multiples of the classical transformations; S_1 and Q_1 are scheme dependent jointly (spin-chain analogue [`n4_untwisted` D2]: one-loop corrections can be moved between Q and S but not removed from both); the scheme-independent combination is the (ξ,η) arity-two relation, whose Δ-part is $\frac{1}{2}(\xi,\eta)$ times the dilatation component (Prop. prop:c2-special, third item). K_1 is tied to D_1, P_1 and M_1 by the (K,P) component as in rem:c7-KDelta; in the Feynman gauge P_1 is a rescaling of the classical translation on the scalar bilinears ($-\frac{1}{3}$ unit in the translation-covariant scheme, design prop:o-P1), and $K_1 \equiv \frac{1}{4} \operatorname{div} v\, \gamma_O O - p \operatorname{ad}_{S^{(1)}(K_b)} O$ for $P_1 = p \operatorname{ad}_{S^{(1)}(P_b)}$ (design thm:o-KD).

6 Open items

1. Normalisation: [BK] (3.64)–(3.69) (Killing-normalised f, factor $\frac{1}{2}$ of footnote 23, propagator (2.31)–(2.32), measure $(2\pi i)^{-2}$) versus [BGKWWY] (C.3)–(C.4) ($SU(2)$, ε_{ABC}, $\frac{1}{4\pi^2}$) versus the project unit; the relation of the twisted $\hbar$ to $\hbar/(X,Y)$ and the normalised field map (3.51) (including the normalisation of $b \leftrightarrow u' F u'$ and $\partial c \leftrightarrow \bar{\lambda} w$) are not given in any source.

2. The constants c_k of ev_ξ: [ESW] Def. 2.17 prints coefficient 1 for $S^{(2)}(Q,Q)$; the signs of the project's cochain values (lem:c2-components, parity $k + \sum |x_i|$) must be matched in code before arity-two numbers are compared.

3. Label identification of Q_-^4, $\pm$, $\dot{\alpha}$ and phases ([CY13] (2.5) $-i$; [GGKM08] (2.6); [KKLP] (2.3); [KLLS] (2.2); [BK] (3.49)–(3.50)) with (u, u', w): `step2/notation`.

4. Representative of b: field ($u'Fu'$, arity two for (3.64)) or antifield ([ESW] Thm. 10.9, arity one); both must give the same class.

BGKWWY App. C internal factor 2 and sign (CK03, CK04), (5.21) sign versus CK02; stress tensor normalised $\operatorname{Tr} b\partial c$ (3.19) versus $2\operatorname{Tr} b\partial c$ (App. B.1).

5. Possible tension (`sweep.json`): [Z05] (2.6) $[\mathfrak{L}, \mathfrak{T}^{\pm}] = \pm\mathfrak{T}^{\pm}$ gives the psu$(1|1)^2$ supercharges only odd powers of g; if $\overleftarrow{\mathfrak{T}}^{+} = \mathfrak{Q}^{1}{}_{1}$ is Q^{4}_{-} up to R-relabelling, the twisted Q_1 restricted to the su(1,1|2) sector must vanish. Neither identification is made in a source. [CLW] footnote 5 reads [Z05]'s leading Q_1 as being of the form $[Q_0, -]$, for the HT supercharge Q_{HT} (Target 12), which is consistent with the $[Q_0, -]$ form and not with a vanishing Q_1 in every scheme.

CL25 calls $C(R)$ "dual Coxeter number of R"; [EJSS05] uses $\operatorname{Tr}(T^a T^b) = \delta^{ab}$, $K = -\frac{1}{3}\operatorname{Tr}$ and $\bar{D}^2 = -\frac{1}{4}\bar{D}\bar{D}$; the identification "Konishi anomaly coefficient $= -1\times$unit" is withdrawn (Target 5 (c)).

6. Not read: arXiv:2401.08113, 1906.04221, 2606.05314, 0905.3738; Zwiebel's thesis (ref. [24] of [Z11]); Konishi 1984, Konishi–Shizuya 1985 (not on arXiv); hep-th/0606087, hep-th/0409009; [BK] (3.70) not re-derived. Twisted-side sources not read for their results: Bomans–Wu arXiv:2311.04304 ([BK] ref. [28]) and Saberi–Williams arXiv:1910.04120 ([BK] ref. [27], the source of the realisation [BK] (3.55)–(3.56); used in this program only for the superconformal algebra, [SW]). [Z11] read for §2.3; [BMV] read for its main result.

7. Statements of [CL22] footnote 10 and [EJSS05] rest on DRED/SSDR and are quoted as statements only.

8. Not re-verified against the sources in the revision of 2026-10-28 (kept as transcribed by the reader files): [BGKWWY] (3.24), (3.26), (3.33), (3.40)–(3.45), (C.1)–(C.2), (C.5)–(C.9), (D.13)–(D.15), App. E, footnote 28; [GKW] (2.27), (2.31), footnote 13; [W18] Thm. 3.4; [BGKWY22] (B.6); [BGWW] (3.46)–(3.52), (4.82)–(4.85); [ESW] Prop. 2.21, Rem. 1.29, §10.3; [BK] App. B; [B03] Table 2.